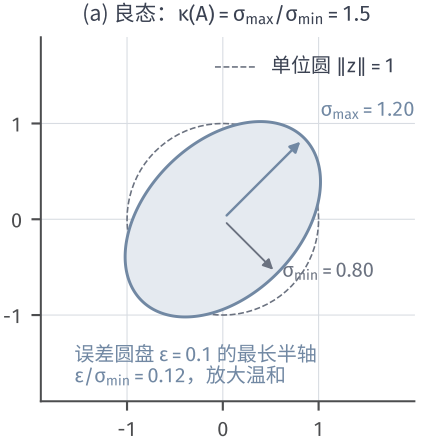
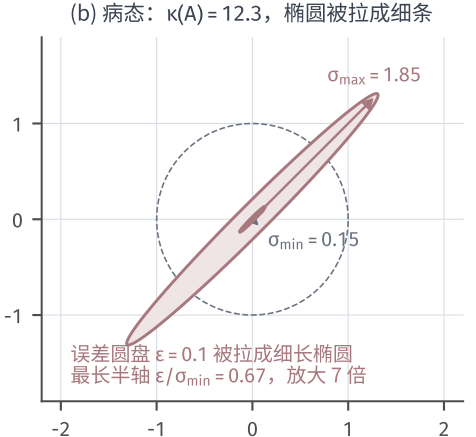

(a) 良态：κ(A) = σmax / σmin = 1.5
单位圆 ‖z‖ = 1
σmax = 1.20
σmin = 0.80
误差圆盘 ε = 0.1 的最长半轴
ε/σmin = 0.12，放大温和
1
0
-1
-1
0
1


(b) 病态：κ(A) = 12.3，椭圆被拉成细条
σmax = 1.85
σmin = 0.15
误差圆盘 ε = 0.1 被拉成细长椭圆
最长半轴 ε/σmin = 0.67，放大 7 倍
1
0
-1
-2
-1
0
1
2